

Sampling Method 1: Hard-First Two-Dimensional Tail Intersection

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Abstract

This document details Sampling Method 1 (referred to internally as *Hard-First Sampling*), a targeted data selection strategy for Approximate Nearest Neighbor (ANN) index tuning. Rather than relying on pure uniform random sampling, which often underrepresents the difficult long-tail queries, Method 1 utilizes computationally cheap probes to map a large initial pool onto a two-dimensional performance plane. It explicitly derives thresholds from the Cumulative Distribution Functions (CDF) of coefficient of variation and convergence, harvesting the intersection of the worst-performing tails (bottom 5%) before backfilling the remaining budget uniformly.

1 Theoretical Formulation

Let $\mathcal{D}$ represent the full dataset of vectors. To construct a representative yet challenging query sample $\mathcal{S}$ of size N_{sample} , we avoid the statistical dilution of hard queries by mining a much larger random pool $\mathcal{P} \subset \mathcal{D}$ of size N_{pool} (where $N_{pool} \gg N_{sample}$).

1.1 Cheap Algorithmic Probes

For every query $q_i \in \mathcal{P}$, we execute a fast, low-budget adaptive search probe. Each probe yields a tuple of performance metrics:

1. **Coefficient of Variation (CV)**: A dynamic internal confidence score indicating search difficulty.
2. **Convergence**: The degree of search convergence achieved under the low-budget probe.

Thus, every query is mapped to a 2D coordinate $(cv_i, conv_i)$.

1.2 CDF-Based Tail Sorting and Intersection

To identify the true worst-case queries without bias towards a single metric, we sort the pool independently along both axes. We formulate the empirical Cumulative Distribution Function (CDF) for both metrics. For a given value x , the CDF $F(x)$ represents the fraction of queries in the pool scoring less than or equal to x .

We define the bottom 5% percentile thresholds by solving for the inverse CDF at $p = 0.05$:

$$cv^{(5\%)} = F_{cv}^{-1}(0.05) \quad (1)$$

$$conv^{(5\%)} = F_{conv}^{-1}(0.05) \quad (2)$$

The hard query set $\mathcal{H}$ is strictly defined as the *intersection* of both bottom-5% tails:

$$\mathcal{H} = \{q_i \in \mathcal{P} \mid cv_i \leq cv^{(5\%)} \wedge conv_i \leq conv^{(5\%)}\} \quad (3)$$

1.3 Sample Construction

The final sample $\mathcal{S}$ deterministically includes all identified hard queries. If the target budget N_{sample} is not yet met, the remainder is populated by sampling uniformly without replacement from the “easy” pool:

$$\mathcal{S} = \mathcal{H} \cup \text{UniformRandom}(\mathcal{P} \setminus \mathcal{H}, \max(0, N_{sample} - |\mathcal{H}|)) \quad (4)$$

2 Algorithmic Visualization

The figures below illustrate the two-step geometric logic of Method 1. Figure 1 explicitly shows the 1D threshold derivation, while Figure 3 maps the 2D intersection dynamics.

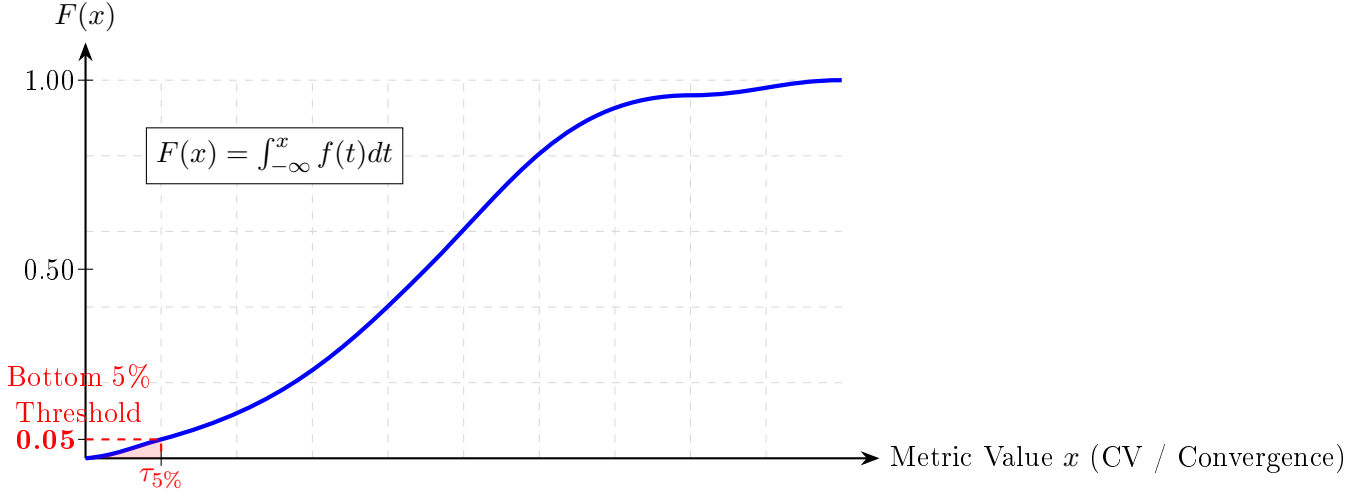


Figure 1: Derivation of the 5% hardship threshold via the empirical Cumulative Distribution Function (CDF). The threshold $\tau_{5\%}$ explicitly isolates the hardest queries on a single axis.

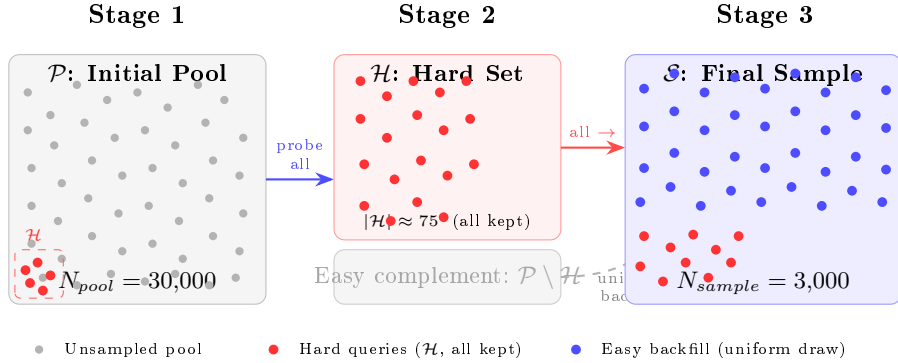


Figure 2: Three-stage Hard-First Sampling pipeline. **Stage 1:** A pool of $N_{pool} = 30,000$ random vectors is probed cheaply to compute CV and Convergence for every candidate. **Stage 2:** The intersection of the bottom-5% tails is extracted as the Hard Set $\mathcal{H}$ (≈ 75 queries, all unconditionally retained); the remaining $\mathcal{P} \setminus \mathcal{H}$ forms the easy complement available for backfill. **Stage 3:** All hard queries are emitted first (red); the budget is then topped up by drawing uniformly from the easy complement (blue) until $N_{sample} = 3,000$ is reached.

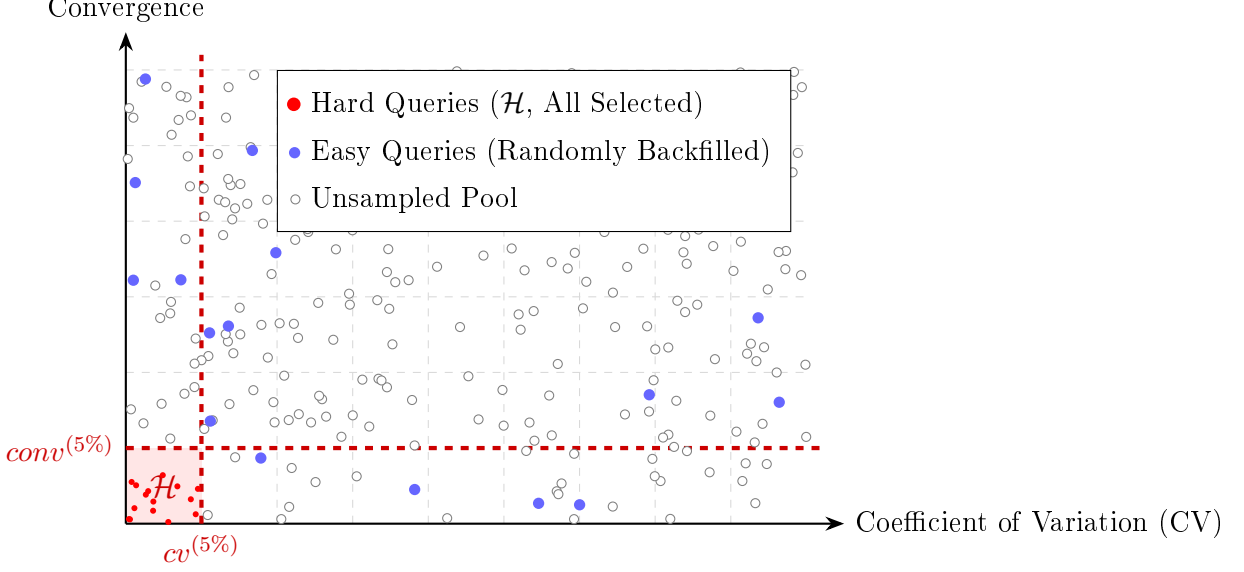


Figure 3: Two-dimensional metric intersection. The dashed lines represent the independent 5% thresholds. Red points residing in the bottom-left quadrant ($\mathcal{H}$) strictly satisfy both hardship criteria and are prioritized. The blue points illustrate the uniform backfill sampling to reach the target budget N_{sample} .

3 Methodological Advantages

The implementation of Sampling Method 1 offers significant empirical advantages for index parameter tuning:

- **Tail Stability:** By deliberately over-representing the bottom 5% of topological difficulties, the optimization target heavily penalizes index structures that fail on edge cases, driving up tail-recall metrics (e.g., $\pi_{0.01}$).
- **Dimensional Independence:** Taking the *intersection* of CV and Convergence (rather than relying on a blended heuristic) prevents false positives where a query has a low CV score simply due to local density variations but is geometrically trivial (yielding high Convergence).